

Homework 11

Please reread Section 1.3 through the end of Example 1.7. This is where Durrett first mentions conditional probability.

Please also read Section 3.1 through the end of Example 3.7.

1. On Midterm 1, you thought about a betting game played by the Earl of Yarborough. In each round of the game, the earl has a probability 0.000547 of losing.
 - a. Suppose the earl plans to play the game until he first loses. Let X be the random variable representing the number of times the earl plays (including his first loss). What distribution should we use for X ?

$X \sim \text{geometric}(0.000547)$

The random time until first “success” for independent trials with probability p follows the $\text{geometric}(p)$ distribution.

- b. Use formulas for this distribution to calculate the expected number of games the earl plays (including his first loss) and the standard deviation for this random variable.

$$E[X] = \frac{1}{0.000547} \approx 1828.15 \text{ games}$$

$$\sigma(X) = \sqrt{\text{Var}(X)} = \sqrt{\frac{1-p}{p^2}} \approx 1,827.7 \text{ games}$$

2. A friend flips two coins and tells you the at least one is a head. Given this information, what the probability that first flip is a head? You will answer this question systematically following the steps below.
 - a. This question is about the chance experiment of flipping a coin twice. Write a suitable sample space with equally likely outcomes for analyzing the experiment. Your sample space should be a set of outcomes and not a number. The sample space should be suitable for answering any questions about flipping two coins. We are not yet thinking about conditional probability.

We can use the set of all sequences of two H/Ts as our sample space: $\Omega = \{HH, HT, TH, TT\}$.

- b. Let A be the event “at least one flip is a head.” Let B be the event “the first flip is a head.” Write the outcomes in the following events. You need to give sets of outcomes and not numbers to answer this question.

$$A = \{HH, HT, TH\}$$

$$B = \{HH, HT\}$$

$$A \cap B = \{HH, HT\}$$

c. Calculate $P(A \vee B)$ and $P(B \vee A)$. Which probability answers the question?

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{4}}{\frac{2}{4}} = 1$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{2}{4}}{\frac{3}{4}} = \frac{2}{3}$$

Which of these conditional probabilities answers the question?

The second conditional probability, $P(B|A) = 2/3$, answers the question.

d. Are A and B independent? Explain your answer using your calculations.

A and B are not independent events. For independent events, we have $P(A|B) = P(A)$ and $P(B|A) = P(B)$. Neither of those equations is true in this case, and so A and B are not independent.

3. A friend rolls two dice and tells you there is at least one 6. What is the probability that the sum is at least 9? You will work out the steps to answer that question below.

a. As in class, we use the familiar sample space from Example 1.2 to model this chance experiment. What events do you need to consider to solve the problem? Describe the events and write the notation for the conditional probability that you need to calculate in terms of your events. Save the calculations for later.

One event is A = "the friend rolls at least one 6" and the other is B = "the sum is at least 9." The conditional probability that we want to calculate is $P(B|A)$.

b. Calculate the conditional probability using the events that you identified in part a. Explain your calculations by listing the outcomes in each event whose probability you need to calculate.

$$A = \{(6,1), (6,2), (6,3), (6,4), (6,5), (6,6), (5,6), (4,6), (3,6), (2,6), (1,6)\}$$

$$B = \{(3,6), (4,5), (4,6), (5,4), (5,5), (5,6), (6,3), (6,4), (6,5), (6,6)\} - \text{we need this event but not } P(B)$$

$$A \cap B = \{(3,6), (4,6), (5,6), (6,3), (6,4), (6,5), (6,6)\}$$

$$P(B \mid A) = P(A \cap B) / P(A) = (7/36) / (11/36) = 7/11.$$

- c. Explain how to reason out the conditional probability using the “reduce the sample space” method.

With this method, to calculate the conditional probability $P(B \mid A)$, we think of A as our sample space. We view all outcomes in A as equally likely, since all outcomes in the original sample space were equally likely. To calculate $P(B \mid A)$, we calculate the outcomes in B that remain in the reduced sample space. Of the 11 outcomes in A , 7 of them come from B , and so $P(B \mid A) = 7 / 11$.

4. Suppose you draw 5 cards out of a deck of 52 and get 2 spades and 3 hearts. What is the probability that the first card drawn was a spade? Let's reason it out.

Let A be the event “the 5 cards you draw include 2 spades and 3 hearts,” and let B be the event “the first card you draw is a spade.” You could view the probability I am asking about as $P(B \mid A)$. It is a bit tricky to calculate that conditional probability using a sample space and the formal definition of conditional probability. (I will show you how to do it in my solutions, but you are welcome to try it yourself.)

Durrett would encourage us to think like a probability person: You drew 2 spades and 3 hearts. Each of those 5 cards is equally likely to have been the first card. What is the probability that the first card drawn was a spade?

Thinking in this way, we view the first card as drawn at random from the 5 cards in our hand. The 5 cards are equally likely to be first, and 2 of them are spades. Therefore, the probability that the first card is a spade is $2/5$.

5. Back to the Earl of Yarborough! In question 4a of the midterm, I asked you to calculate the probability that a hand of 13 cards has no card with value 10 or higher. In my solutions, I used the “urn problem” method to find the probability.

Let's solve a simpler related problem using conditional probability. Instead of drawing 13 cards, imagine drawing 2 cards. Let A be the event “the first card does not have value 10 or higher.” Let B be the event “the second card does not have value 10 or higher.” We will calculate $P(A \cap B)$.

- a. In Section 1.3 and in Section 3.1, you read about the conditional-probability multiplication rule, which is a formula for calculating $P(A \cap B)$ using conditional probability. Write this formula:

$$P(A \cap B) = P(A) P(B \mid A)$$

- b. To use the formula, you need to calculate $P(A)$ and $P(B | A)$. Following the model of Example 3.4 (similar to a class example), calculate these two probabilities and thus calculate $P(A \cap B)$. Explain your thinking (in the style of Example 3.4) for the first two probabilities.

$P(A) = \frac{32}{52}$ There are 52 cards in the deck, and all are equally likely to be the first card. Of those 52, 32 (8 in each suit) have value that is not higher than 10 (2, 3, 4, 5, 6, 7, 8, or 9).

$$P(B|A) = \frac{31}{51}$$

After a card not higher than 10 is drawn, there are 51 cards left in the deck. All 51 are equally likely to be drawn next. Since a card whose value is not higher than 10 was drawn first, of those 51, 31 have value that is not higher than 10.

$$P(A \cap B) = \frac{32}{52} \cdot \frac{31}{51}$$

You could follow the same pattern of reasoning to calculate the probability that the earl does not draw any card with value 10 or higher, giving a new solution to question 4a from the first midterm.

If you followed this same pattern of reasoning, you would get

$$\frac{32}{52} \cdot \frac{31}{51} \cdot \frac{30}{50} \cdot \frac{29}{49} \cdot \frac{28}{48} \cdot \frac{27}{47} \cdot \frac{26}{46} \cdot \frac{25}{45} \cdot \frac{24}{44} \cdot \frac{23}{43} \cdot \frac{22}{42} \cdot \frac{21}{41} \cdot \frac{20}{40}$$

That product works out to be approximately 0.0005470333582, matching what I found in the midterm solutions using the “urn problem” method. (The fractions from the two methods are equal.)